

Grant Miller

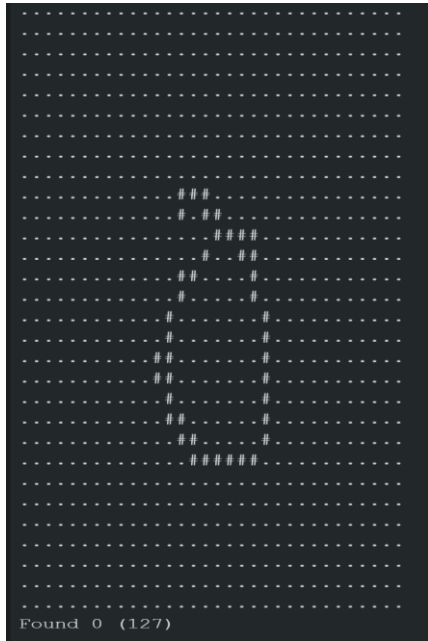
EE 446

15 May 2026

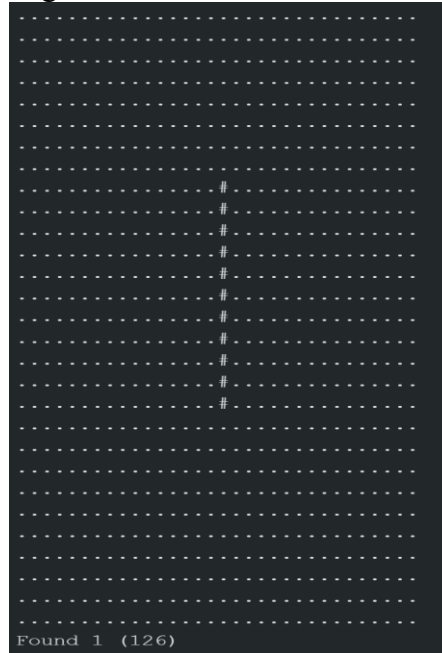
Lab 6 Report

Task 1: Deploy Baseline int8 Quantized Model

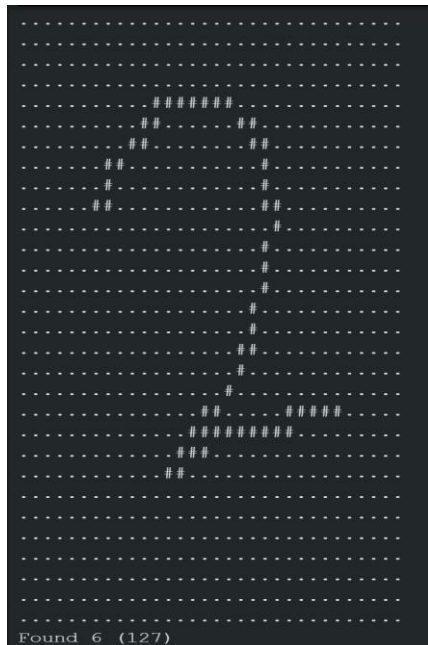
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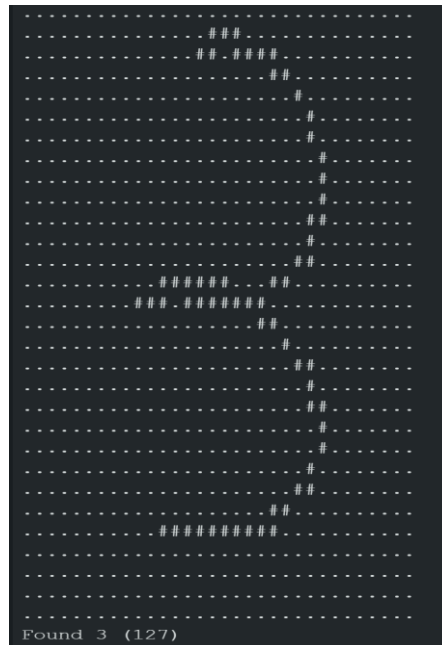
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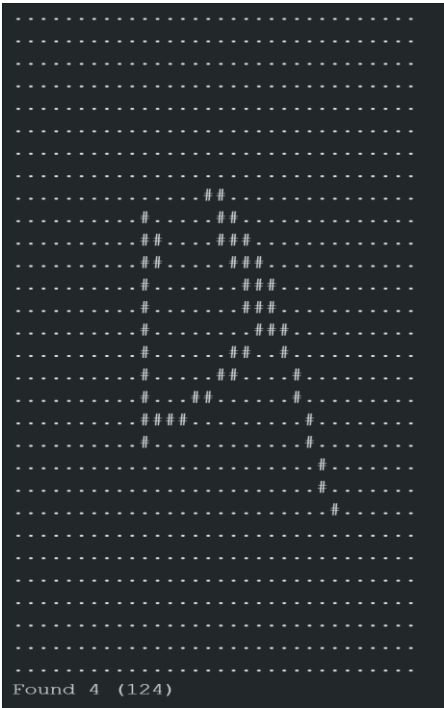
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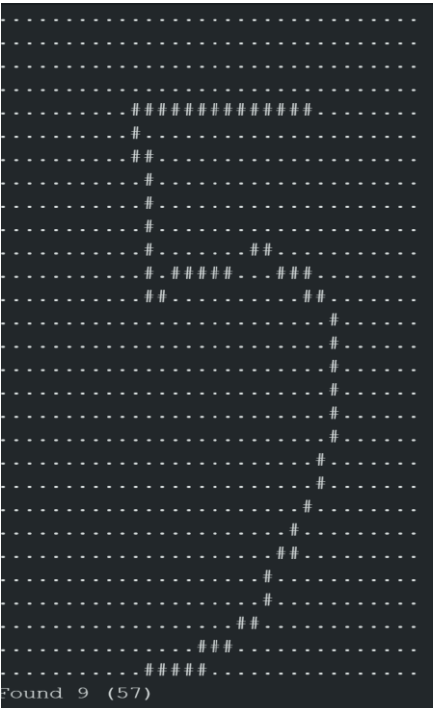
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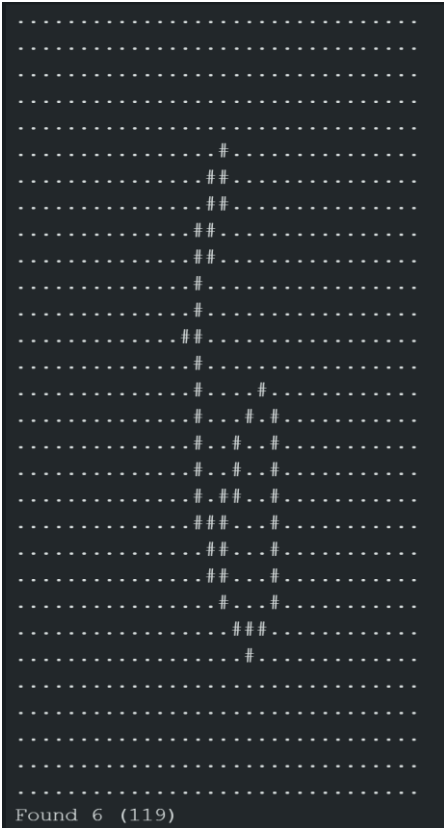
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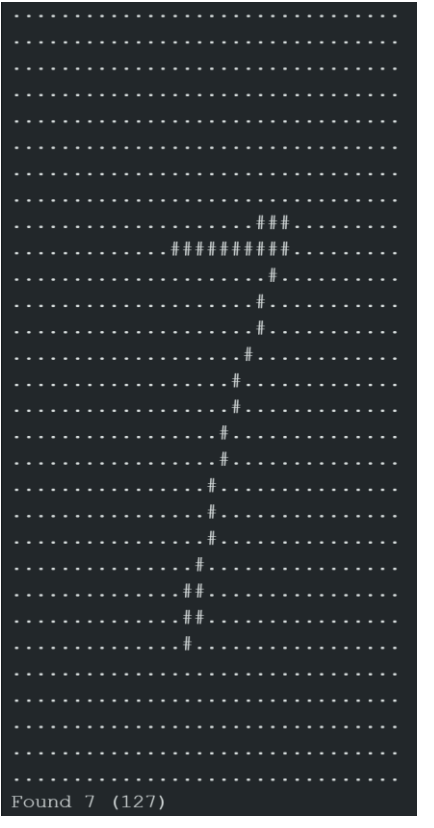
Digit = 5 – looks like unoptimized model struggles with 5



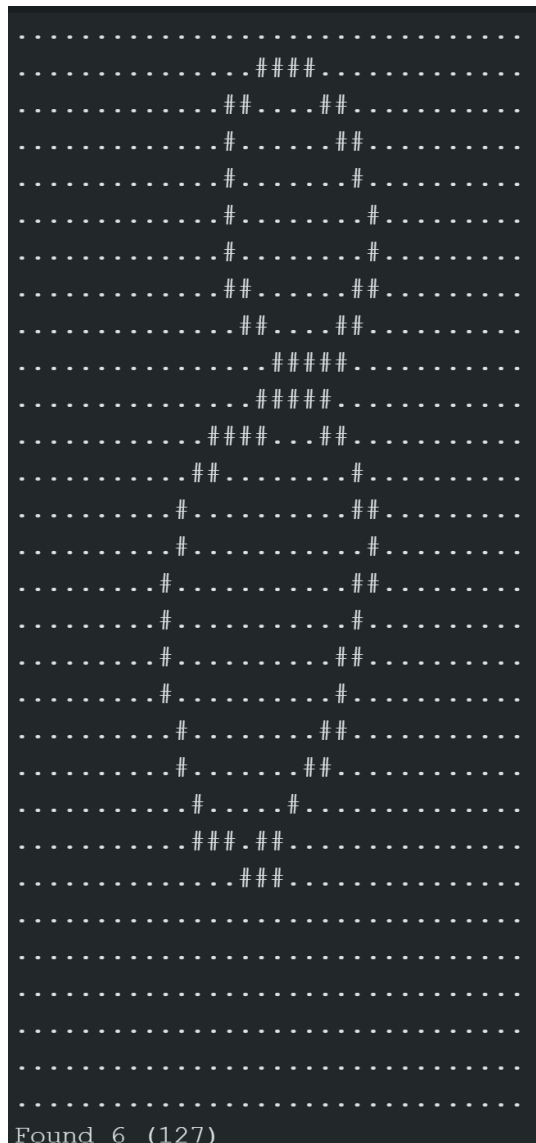
Digit = 6



Digit = 7

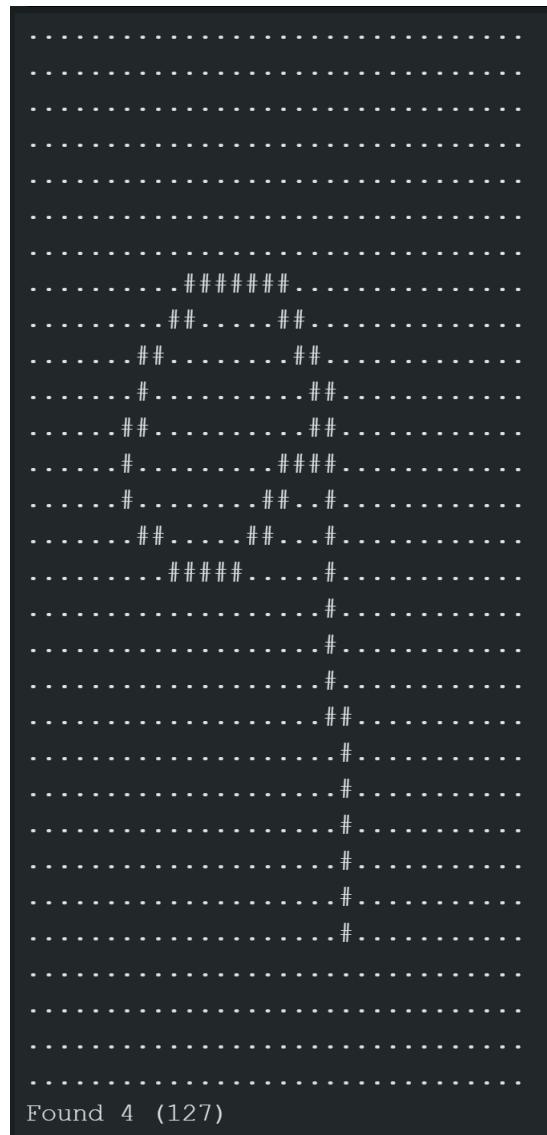


Digit = 8



This model also struggles with 8.

Digit = 9



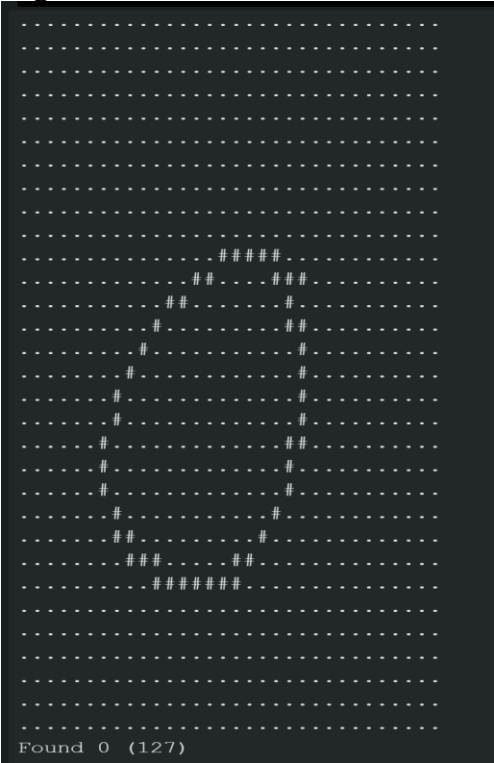
This model also struggles with 9.
Sidenote, this 9 is so pretty.

Task 2: Collecting Data

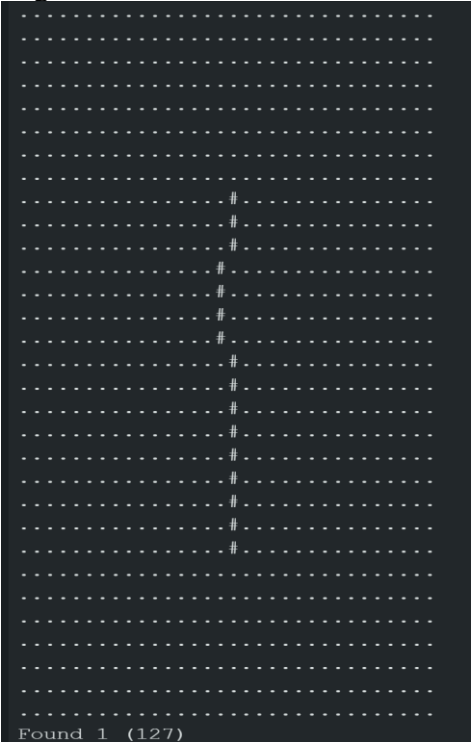
Done! JSON file used in training for the fine-tuned model in task 3.

Task 3: Deploy Fine-Tuned int8 Quantized Model

Digit = 0



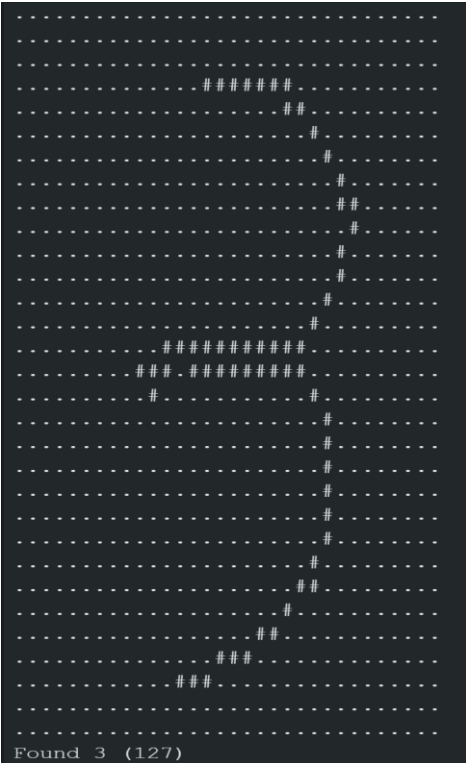
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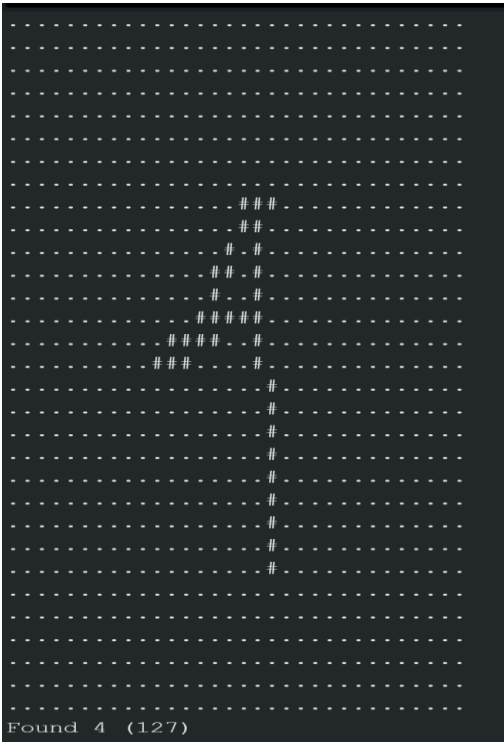
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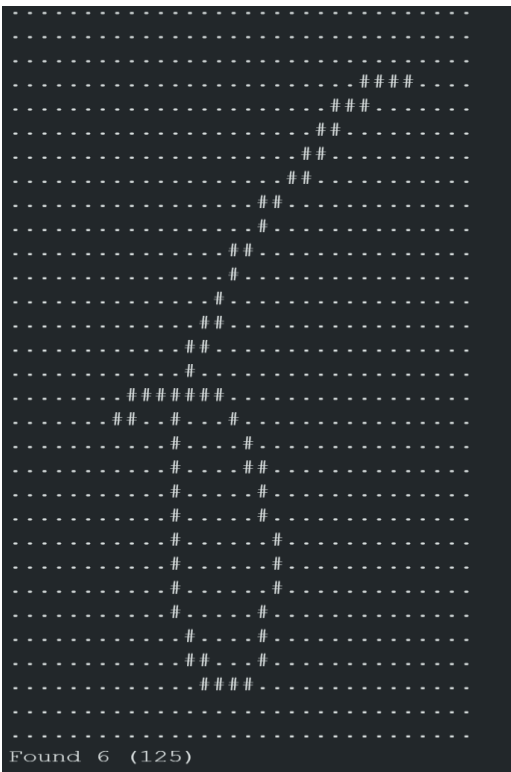
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Digit = 4



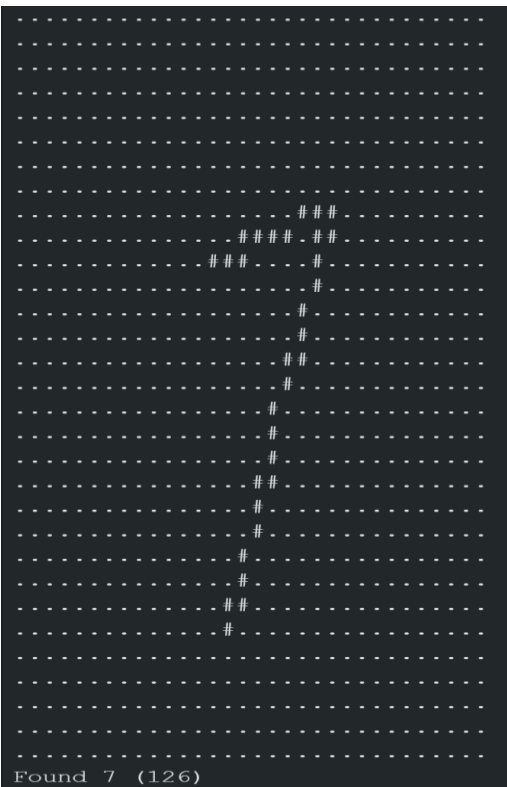
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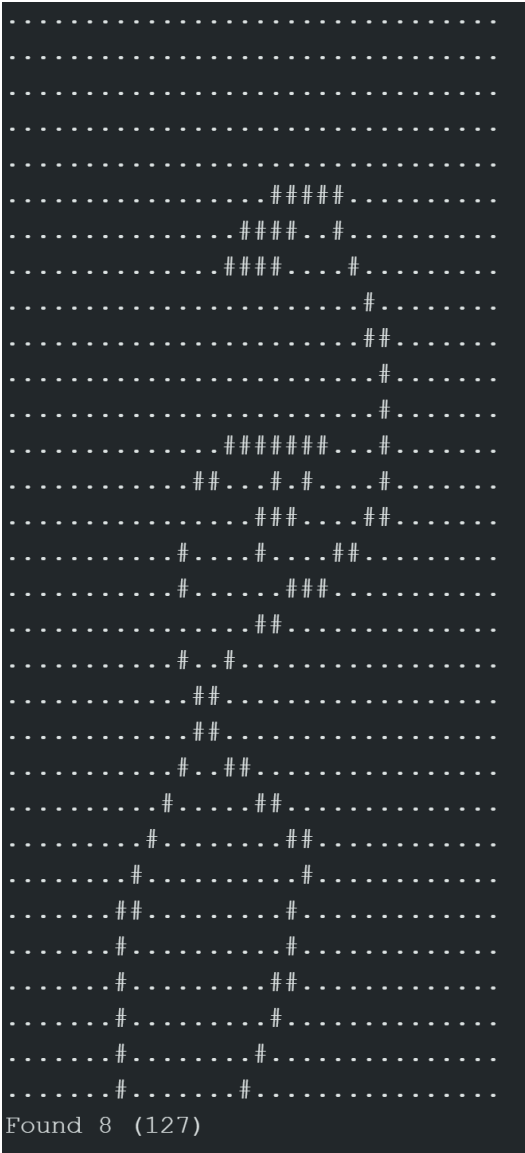
Digit = 5 – looks like unoptimized model struggles with 5



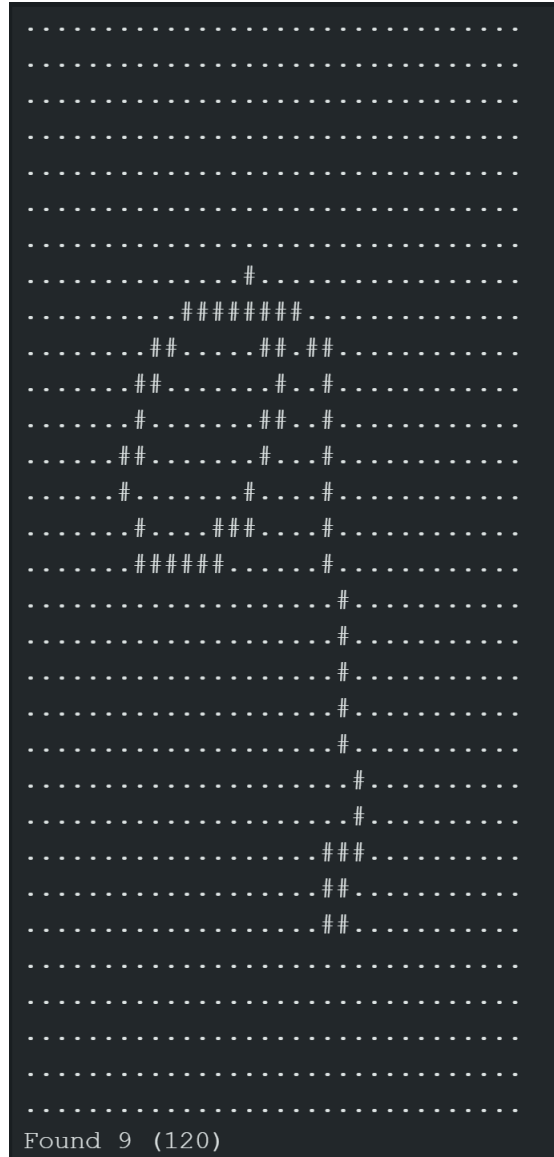
Digit = 7



Digit = 8



Digit = 9



Task 4: Compare Results and Reflect

1. The main difference I observed was the ability for the fine-tuned model to pick up on a different variety of gestures like drawing a 2 starting from the top vs the bottom. It was able to pick up both vs when I did that for the first model it only pickup up on one method of drawing. Also, the fine-tuned model was actually able to identify 5, 8, and 9 while the not fin-tuned model wasn't even when the inference strongly resembled the correct number I was drawing.
2. The fine-tuned model performed better as it was able to pick up on numbers the first model could not. Confidence levels for the first model when it correctly identified a

number were comparable. Another thing to note is that the fine-tuned model would not have a high confidence level when the inference was scribbly nonsense. I noticed sometimes that the not fine-tuned model would still have high confidence when the drawing did not resemble anything.

3. Yes, fine-tuning helped for my specific gesture style. I draw numbers in a specific way and that was reflected in the training data which was then reflected in the model.